

SETI FALSE ALERTS AS "LABORATORY" TESTS FOR AN INTERNATIONAL PROTOCOL FORMULATION

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Abstract---The recent advances made in R&D on an efficient multi-million channel radio receiver constitute a major step in SETI. With the prospect of better chances to detect a signal of artificial extraterrestrial origin, it becomes important to work on the formulation of an international protocol for subsequent activities. On the other hand, a few professional astronomers have had the thrilling experience of living such an extraterrestrial adventure when they accidentally fell upon plausible candidate signals. The best known example was provided by the discovery of pulsars some 20 years ago. I suggest that the study of these experiences, though based on false alerts, could contribute to the formulation of the protocol through the "laboratory" tests they provide. At the International Astronomical Union Colloquium 99, I described my own experience. Here, taking advantage of my radioastronomical background, I shall describe the CTA 102 affair initiated by Kardashev in 1963.

1. INTRODUCTION

The scientific community is becoming aware that in case a success is achieved in its efforts to discover an extraterrestrial intelligence signal, the question of how to make known such an important discovery will present a delicate problem. A strong tendency is developing for the preparation of an international protocol[1]. Since SETI (Search for ExtraTerrestrial Intelligence) has started to become a scientifically realistic endeavor[2], a few astronomers have had the thrilling experience of false (in the end!) alerts. These live events provide valuable "laboratory" tests for the development of such a protocol. At the very least, it should be noticed right away that they show how carefully the observational facts should be assessed in order to increase to the utmost level the degree of certainty of an eventual discovery. Indeed, one should not forget that when in science we look for an expected result from the use of new instrumentation we rarely get a yes/no answer; most often we are faced with a new puzzle.

In this paper we shall restrict ourselves to the case of reception of a radio signal, which is the most advanced form of SETI now. Already in this particular radio context there have been a number of live false alerts. Maybe the oldest dates back to August 1924 when D. Todd, after asking for all U.S. transmitters to be silent, listened at 5-6 km wavelength radio during the closest approach of the planet Mars and got only a potpourri, not ascribable to an extraterrestrial source[3].

Next we have the famous Ozma search when F. D. Drake listened with the new 85 foot radiotelescope at Green Bank to two close solar-type stars and got a tremendous "Wham!" on the recorder in April 1960[4].

In the Summer of 1967 S. J. Burnell, observing radio sources with the Cambridge interferometer, "became aware that on occasions there was a bit of 'scruff' on the records." Little green men sent regular pulses? No, it was pulsars, a major discovery[5].

Then, during the longest search of all, with the Ohio State University large radio telescope, when R. S. Dixon was shown on the records a "Wow!" written next to a big radio signal in the margin of the print-out by his collaborator, the amazement was at its top level[6].

And there was my own false alert, which rambled along on the outskirts of my work on galaxies for six years, with its ups and downs, pointing to an anonymous faint solar-type star hundreds of light-years away.

Here I shall report on a soviet experience, the "CTA 102 affair," which N. S. Kardashev started in December 1963 and which, for quite a number of years, aroused a major interest in the scientific community. Contrary to my own experience, which was lived from the inside, here I shall take the affair from outside, already a quarter of century later. I shall reconstruct it from the scientific papers which have been published about it, acting as an external scientific observer. I shall not refer to the reports which appeared in the press media. I regret that I could not discuss this with Kardashev himself, because of lack of time, before submitting this external view. I would welcome any personal comments which Kardashev might wish to produce on his experience; they would be most enlightening.

2. THE ORIGINAL PAPER

I shall start from the original scientific paper which Kardashev submitted to the *Astronomicheskii Zhurnal* on December 12,

1963 under the title "The Communication Of Information By Civilizations On Other Worlds[8]."

He was then a 31 year old scientist at the Sternberg Astronomical Institute in Moscow. His paper was quite a seminal one, in which new concepts were treated, most of them for the first time. The paper aimed to investigate the conditions required to obtain information transmission at the maximum rate possible across space with electromagnetic waves.

The first condition is that all sources of noise entering the reception measurements be reduced. One is the receiver noise which, rightly now, Kardashev considered as negligible. Then there is the one created by the sky background where the instrument is pointing; it comes from thermal and synchrotron emission produced by the various celestial bodies in the universe; from our solar system vantage point it is maximum along the Milky Way plane, peaking towards the galactic center. Last there is a statistical noise arising from the quantum corpuscular nature of electromagnetic waves, especially important at short wavelengths.

The overall situation was depicted by Kardashev in a graph where the noise is evaluated, as usual in radioastronomy, as an effective temperature (which is linked to the flux density per steradian through the Planck equation for black body radiation). In this graph, there is a strong minimum in the decimetric/centrimetric range, which is thus the most suitable one. Note that at the time the paper was written, the 3K cosmological background was not yet discovered and that it adds its 3 degree contribution all along the graph.

Now the information transmission rate depends on the power available to the emitting civilization. In order to obtain the maximum rate, which is Kardashev's major point, one has to solve a variational problem using Shannon's theorem of information theory. The result is very simple: the spectrum (i.e. flux density versus frequency) of the transmitter has to have the same shape as the noise curve, but reversed in sign.

Next Kardashev considered which values of the transmitter power P could be achieved. It is here that he introduces his concept of civilizations of types I, II and III, which respectively could utilize power equal to the power now used on this planet, the power of their central star, and the power of their own galaxy. This is quite a farsighted view, and Kardashev does not hesitate to argue that type II

and III civilizations are the most likely ones to be discovered in our searches.

With these parameters in hand he finds that with a comfortable signal-to-noise ratio of 100; (a) our present sensitivities allow detection of even a civilization transmitting isotropically, and (b) that a few $\times 10^9$ bits per second could be received from a type II civilization anywhere in our Galaxy or from a type III civilization anywhere in the observable universe.

The powers being discussed here are comparable to the powers of supernovae and radiogalaxies. It is thus necessary to find criteria to distinguish artificial emissions from natural ones. Kardashev lists the following:

- the shape of the spectrum with the maximum already mentioned;
- an angular diameter smaller than a milliarcsecond (in view of the modulation and distances implied);
- a circular polarization, which is not modified as it propagates through interstellar space;
- a variability at a short time scale for information transmission;
- some evident signature for artificiality, such as a gap around the neutral hydrogen 21 cm wavelength, or an inverse frequency dependence of the emitted flux density at high frequency as a result of the noise spectrum.

Then he looks at the observational status for the epoch of his paper: 20-30 radio sources were known with diameters smaller than 1-10 arcsec, some of which are not identified with peculiar optical objects. Among them CTA 21 and CTA 102, "even more intriguing, exhibit a spectrum highly similar to the anticipated artificial spectrum."

Thus was started the CTA 102 affair.

3. THE ASTROPHYSICAL CONTEXT

This affair happened in the midst of a very important stage of the development of astrophysics: the discovery of quasars. Since in the end CTA 102 turned out to be a quasar, it is very interesting to consider the astrophysical context thus created.

The first identification of a radio source as a "quasi-stellar" object was made in 1960[9]. The Owens Valley radio interferometer had given radio positions to within 30 arcsec, while the Jodrell Bank telescope had measured diameters down to 1 arcsec. Among the radio sources measured was 3C 48 (3C is for the third Cambridge catalogue, while CTA is for the first Cal Tech one). A plate was taken with the Palomar 5 m telescope on September 26, 1960,

with every expectation of finding a distant cluster of galaxies, but measurement of the plate gave the unexpected result that the only object lying within the error rectangle of the radio position was one which appeared to be stellar. It was associated with an exceedingly faint wisp of nebulosity whose peculiarity made the identification almost certain.

It took several years to realize that such star-like objects were not in our Galaxy. Schmidt, in March 1963, identified another one, 3C 273, and successfully deciphered its spectrum, discovering that it had a colossal redshift, which put it far out in the realm of extragalactic space. The implication was that it was intrinsically 100 times brighter than a bright galaxy; the new branch of quasars was born[10].

In December 1963, at the moment Kardashev submitted his original paper, an historic conference was being held in Dallas[11] to deal with the problems of ultra-high energies produced in radiogalaxies. 400 scientists, among them top ones like Hoyle, Oppenheimer, Fowler, the Burbidges, Morrison, Wheeler and Dyson, discussed new ideas for possible sources of energies reaching 10^{57} - 10^{61} ergs. Schmidt presented his quasar results where 3C 273 sat comfortably among radiogalaxies with an energy output of 10^{59} ergs.

Kardashev did not attend and his ideas did not appear there. A general trend was developed to explain the emission through synchrotron radiation of relativistic electrons in magnetic fields, where the electron energies would be provided by the gravitational collapse of some massive central body. Note that in a 1969 review the Burbidges wrote that "after nine years, quasars are still puzzling astronomers[12]"; even presently the central engine is still a kind of black box!

4. OBSERVATIONAL DEVELOPMENTS FOR CTA 102

We now look at the developments concerning Kardashev's CTA sources that took place in different fields, with ups and downs in the suspense they created. Let us listen to Pariiskii in 1964: "To satisfy N. S. Kardashev's inquiries, we measured the integral flux of CTA 21 and 102 in the centimeter wavelength region using the Pulkovo large radiotelescope at 6.4 cm....The high frequency points follow fairly closely a linear flux versus wavelength dependence. This is the expected dependence for artificial radio spectra[13]." Quite a treat! But Pariiskii strikes

a cautious note: "This spectrum, however, is characteristic of other objects as well and it can be explained within the framework of the synchrotron radiation mechanism." This is an echo to the Dallas conference of the previous December. Other encouragements came from Sholomitskii, also from the Sternberg Institute, who monitored CTA 21 and 102 from August 1964 to February 1965 and submitted his results for publication on February 2, 1965[14]. At 32.5 cm CTA 21 was constant but CTA 102 exhibited a sinusoidal 30% variation with a period close to 100 days! Furthermore, using the physics of synchrotron self-absorption[15] to determine a 0.01 arcsec angular diameter for the emitting plasma and using the distance travelled by waves in 100 days as an upper limit for its size, he found that the source was likely to be in our Galaxy.

This remarkable variability incited other workers to check it. Within two years measurements were analyzed at 168, 31.3, 25.5, 21.0 and 13.1 cm with no significant evidence of variations! However, Sholomitskii went on with his monitoring and on March 5, 1966[16] he published a remarkable curve confirming his previous results. What can be concluded from that? I think the situation is well summarized by Shakeshaft in his 1967 report: "The first quasi-stellar source reported as variable was CTA 102 by Sholomitskii from observations at 32.5 cm. His later results indicate a very regular fluctuation of intensity of a circularly polarized component with a period of 102 days. No other observers have detected any variation[17]." The last remark about frequency and polarization is very important, especially in view of Kardashev's criteria.

At Pulkovo, Pariiskii was also able to reduce the size of the rectangle error of the radio position to the point where he could identify the source with a very faint bluish stellar object[13]. At the same time, pursuing the road opened by the identification of 3C 48, California astronomers also identified CTA 102 with the same "star"; an abnormally blue, ultraviolet-excess stellar object of magnitude 17, qualifying it as the "the most interesting new identification[18]."

Another criterion: the diameter of the object. In the optical range the atmospheric turbulence limits the precision to one arcsec, which is then an upper limit of the angular diameter. In the radio range an upper size 10×12 arcsec was given in 1964 by Wyndham[18] which soon was narrowed to 0.4 arcsec with the Jodrell Bank 132 km interferometer[19] and to 0.2 arcsec with the

Pulkovo radiotelescope[13]. On the contrary, CTA 21 was shown to be extended and can be disregarded hereafter.

After these strong encouragements for Kardashev, the next step was theoretical, but not necessarily along his ideas. A natural way to account for the maximum presented by the radio spectrum is to use synchrotron self-absorption: a plasma with relativistic electrons in a magnetic field radiates a synchrotron spectrum which is attenuated at long wavelength because the photons are absorbed inside the plasma. In this way it can be shown that the angular diameter can be evaluated from the flux density and frequency of the maximum in the spectrum[15]. Such calculations applied to CTA 102 give 0.01 arcsec. This was then among the smallest angular diameter estimates.

Continuing with his discovery of the true nature of quasars by deciphering the spectrum of 3C 273, Schmidt submitted for publication his measurements of the CTA 102 redshift on April 8; a very high value of 1.037. Using relativistic models of the expansion of the Universe, this permitted the determination of the intrinsic properties of this object lost in the depth of space:

- absolute optical magnitude: -27;
- monochromatic flux at 1 GHz: 4×10^{27} W/Hz[20].

All these values are typical of quasi-stellar radio sources. This was, for all practical purpose, the *coup de grace* for the artificial nature of CTA 102.

5. THEORETICAL DEVELOPMENTS

As Schmidt's measurement occurred almost two years after Kardashev's original paper, it is interesting to follow up the ideas and theoretical developments which Kardashev pursued meanwhile, so much so that Sholomitskii's observations of oscillations kept going on. For this purpose we may use the proceedings of a very important SETI meeting which was held in Byurakan in May 1964: the First All-(Soviet) Union Conference on Extraterrestrial Civilizations[21]. In a session of V. A. Ambartsumyan, Kardashev delivered a paper with the title "Transmission Of Information By Extraterrestrial Civilizations."

In addition I shall also use a scientific book published in 1969 in Moscow under the editorship of S. A. Kaplan, which according to its editor, "is the first scientific monograph in the literature on the subject of the search for signals

from extraterrestrial civilizations[22]." It contains an important contribution from Kardashev: "The Astrophysical Aspects of the Search for Signals from Extraterrestrial Civilizations."

He starts by stating what he calls the main dilemma, according to which "there is a high probability that civilization is a universal phenomenon, and yet there are no currently observed signs of cosmic activity of intelligent creatures." This is a weak form of the so-called Fermi paradox. To this he sees two alternatives:

- current data on absence of civilizations is wrong;
- some fundamental factor slows down the development of civilizations.

This leads him to review the available astrophysical data, from the formation of stars, heavy elements, and the solar system, up to the then recently discovered quasars. His main conclusion is that civilizations could have come into existence already billions of years ahead of us.

Thus, he pleads for a necessary observational exploration of all of the electromagnetic spectrum, with a harvest of at least a hundred objects in each domain. We are very far away from that and he concludes "that the entire dilemma is most probably a product of insufficient knowledge on our part."

To pick up astrophysical signs of civilizations, we must consider their main features and their development. He bypasses the definition of life and sees a civilization as a highly stable state of matter mainly supported by information manipulation.

This allows for an unlimited development. Citing a U.S. report on controlled fusion, he recalls there is a definite limit on Earth imposed by climatic considerations, which with the present annual growth will be reached in only 125 years. If such a growth could be sustained, within 1,500 years the human power output would reach the total radiation output of the entire galaxy.

The population also grows exponentially and within 2,000 years would reach the mass of ten million galaxies. And the quantity of information would reach, in the same time, a number of bits exceeding the number of atoms in the observable universe.

Kardashev's inevitable conclusion is that the current exponential growth rate constitutes a transient phase and will be unavoidably restrained. Already, if one considers the mean mass density of some space medium to harness, the velocity of light will limit the rate of growth

to a t-squared law, and this within the next millennium.

Hence his main point: there is virtually no civilization in our intermediate stage of development. However, as already pointed out by Shklovskii: "The exponential development of terrestrial civilization suggests possible intelligent activity on the cosmic scale. And the existence of such supercivilizations is of decisive significance for interstellar radio communications, as is obvious from Kardashev's work (approximate quote)."

What civilizations, then? Kardashev suggests the following division into three types, as already mentioned in his original paper:

- I. technological level close to the present-day level on Earth, at 4×10^{19} erg/sec;
- II. civilizations which have harnessed the energy output of their star, at 4×10^{33} erg/sec;
- III. civilizations which have harnessed the energy output of their galaxy, at 4×10^{44} erg/sec.

It takes a few billion years for a type I civilization to develop, no more than a few thousand years for the transition from I to II, and a few tens of millions of years for the transition from II to III.

Given the age of galaxies, one should look for a few type II civilizations in our Galaxy and for type III ones in some other galaxies. Contrary to the present trend, Kardashev resolutely turns his views toward the activity of supercivilizations.

One of the most probable elements of this activity should be the transmission and exchange of information; we should look for them. In his original paper Kardashev had as a leitmotif the determination of the best conditions for obtaining the maximum information transmission rate. In Kaplan's book he is more open and considers, in addition, the determination of the best conditions for detection of a call signal.

He refines the original mathematical formulations, giving explicit, more general equations which, in some limiting cases, fall back on the previous simpler forms based on Shannon's theorem. For an ideal antenna, the rate of information transfer C in bits per second per hertz is a function of the emitted power spectrum at the receiver input P and of the effective temperature of background noise at the input T . For each frequency f of the waves used, $C = C(r, s)$ where $r = hf/kT$ and $s = P/kT$.

In the classical case $r \ll 1$, C reduces to the Shannon case $C = \ln(1+q)$ where $q = P/hf$ is

the number of quanta involved per unit time. In the quantum case $r \gg 1$, $C = \ln(1+q) + q \ln(1-1/q)$, which, in case of high intensities $q \gg 1$, reduces also to the Shannon case.

In case of beacon signals, one has to find the best frequency f and the best signal-to-noise ratio S/N for a given total energy flux received F and a total observing time t available to search in frequency and in direction. The noise N depends on the background flux density per steradian I and on the statistical fluctuations in the number of photons. The signal S depends on F and on the effective area of the receiving antenna A .

In the practical classical case the signal-to-noise ratio S/N reduces to $S/N \propto F \sqrt{Aft}/I$. Hence the conclusion for beacon signals:

- No conditions can be placed on the bandwidth used b nor for the integration time t separately, just a large $t (= bt')$;
- Use decimetric wavelengths 10-50 cm where the background flux density per steradian I is minimum.

Now Kardashev turns to his favorite case: as in his original paper, it is a variational problem, making use of the detailed information transfer rate C above, with F and I given. One has to find the shape of the emitted spectrum E versus f such that the rate of information is maximum. It turns out that when neglecting quantum fluctuations of the signal: $E + I \lambda^2 / A = a$ a function K of r , hf/A and a , where λ is the wavelength and a is some shape factor.

From the fact that E is a positive quantity, one deduces $T < A/2ak$, a condition which fixes the range of frequencies where the emitted spectrum sits. The result is that the optimum transmission range corresponds to the region where the effective background temperature is minimum. In fact, it is the region where the cosmological 3K background prevails, i.e. from submillimeter to decimeter wavelengths; a very wide range of frequencies.

Having found the range, the second step is to find the optimum shape of the transmitted spectrum E . This depends essentially on the dependence of the collecting area A versus frequency. Kardashev considers two cases:

1. If antennas with a beam size L are independent of f , e.g. dishes whose manufacturing accuracies are proportional to their diameters, then $A \propto \lambda^{-2}$. For the case when F is small the function K has a plateau, and thus $E = a$ constant, IL . Then the maximum of the emitted spectrum lies at the minimum of the background flux density per steradian, i.e. in the decimetric range, and

thus coincides with the best frequencies for beacon signals.

Furthermore, the shape of the spectrum is the same as the background one, except that it is reversed in sign, a result previously given in the original paper. Here Kardashev recalls that he gave CTA 102 as an example in 1963.

2. antennas with a collecting area A independent of frequency, e.g. the case of using a unique dish. Again, for small F , the function K is more complex and the result is that for $r < 1$, E has a plateau at long wavelengths, then reaches a maximum about 3 times stronger for $r \sim 1.1$, i.e. at 5 mm, then drops down sharply. In that case the emitted spectrum is much wider and extends from decimetric to millimetric wavelengths. According to Kardashev "a striking example is 3C 273B," the central part of the quasar 3C 273; with no comments from him.

6. THE SCIENTIFIC ATMOSPHERE

To understand the development of a SETI affair it is useful to investigate the scientific atmosphere. From the published scientific papers it is already possible to get a preliminary insight into the scientists' reactions to Kardashev's proposal. I. S. Shklovskii, from the same Sternberg Institute, and Yu N. Pariiskii, from Pulkovo Observatory, explicitly refer to Kardashev's work; the first one in a supporting way, the second in an open-minded manner. However, G. B. Sholomitskii, also from Sternberg, who started to observe CTA21 and 102 the following August, does not refer to him, and among the authors previously mentioned who were checking CTA 102 variability, I failed to see any reference to Kardashev.

More insight may be obtained from the published discussions which were held at the Byurakan May 1964 conference. V. A. Ambartsumyan, Director of the Byurakan Observatory, called attention to the "most paradoxical" part of Kardashev's paper: to send monochromatic, anisotropic, intermittent signals, and to look for continuous, isotropic and uninterrupted signals. It is possible that this remark contributed to Kardashev's later consideration of the beacon signal case.

In the full session discussion, Kardashev and Shklovskii argued for wide band, very powerful, isotropic transmitters that are rare, but detectable from far away. Most of the others, i.e. Ambartsumyan, Zel'dovich, and Martinov,

favoring narrow band, less powerful beamed transmitters. However, they all agree on the centimeter-to-decimeter wavelength range.

In concluding this discussion, Kardashev takes note of this two-sided view and advises a parallel approach: the search for supercivilizations should be an easier task and follow the astrophysical research trend. He recommends a sky survey in centimetric wavelengths with interferometers down to 0.1 arcsec resolution and 0.1 flux unit sensitivity. At the same time Kardashev recommends that more complex equipment should be developed for monochromatic detection from objects within a thousand light-years.

His ideas gave impetus to monitoring the identification of redshift observations. He is humanly a little reserved about the synchrotron radiation explanation for some compact quasars. "It is too early to suggest that quasars are artificial sources, but this hypothesis definitely deserves more than a cursory glance" he says, adding elsewhere that "natural conditions in interstellar space may greatly simplify the creation of ultra-powerful narrow-band radio generators." Quite a far sighted perspective...

7. PUBLIC RELATIONS

As I said in the introduction, my intent was to use only scientific papers. However, in his Russian *Astronomicheskii Tsirkular*, Sholomitskii refers to the Pravda newspaper of April 14, 1965[16]. This is quite an unusual procedure in scientific literature. Thus, I looked for more information in a popular book by Biraud and Ribes[23]. It appears that on April 12 a Tass Agency press release announced that Soviet astronomers had observed signals which might come from extraterrestrial intelligence. On April 14, there was a press conference in Moscow under the chairmanship of Martinov, with Shklovskii, Sholomitskii, and Kardashev, where the CTA 102 case was reported.

This follows by two months Sholomitskii's report on radio variability in the *Information Bulletin*[14], and may have been spurred by this variability. Unfortunately, in this period of time - on April 8 exactly - M. Schmidt submitted to the U.S. *Astrophysical Journal* his results on the high redshift of CTA 102, unveiling its quasar nature[20].

Some may wish to debate the cautiousness of the Soviet scientist, but it is too bad that international scientific exchanges were not more efficient; the question would have been settled

right away. As a matter of fact, I understand from Biraud and Ribes' book that a few days after the press conference in Moscow, a denial was released. But it was too late: *le lievre etait lache!*

8. CTA 102 TODAY

Nowadays, CTA 102 is one among the many quasars accessible to present instruments. Its structure has been investigated down to a resolution of one milliarcsecond by intercontinental radio interferometry. At 18 cm it has two components: a 3 flux unit unresolved source smaller than 2 milliarcseconds; i.e. smaller than 24 light-years, and a larger 2 flux unit component with a 4 milliarcsecond size. The compact core is displaced 10 milliarcseconds to the NNW[24]. It cannot be ruled out that these two components are embedded in a much larger envelope extending to 0.1 arcsec with a brightness of 1-2 flux units[25,26].

CTA 102 was monitored from the years 1966-1971 at 2.8 and 74 cm, showing significant variations of some 10%, but not following Sholomitiskii's behavior, which "may be a relatively common property of discrete radio sources." Because of these variations, it happens that the maximum in the spectrum sometimes disappears[27]. Also, the short wavelength part follows a square root law rather than the predicted linear one[28].

The degree of circular polarization has been found to be very small, on the order of only a tenth of a percent at decimetric wavelengths[29].

9. CONCLUSION

This scientific history of the CTA 102 affair will show, I hope, the kind of puzzle which may face workers in some future SETI adventure. With it in hand, let me strongly support the following resolution: "that before a public action is taken in a SETI affair, it is assessed confidentially by an international interdisciplinary committee of scientists under the responsibility of some of our international scientific unions."

At the same time, I would like to stress that SETI is a scientific research theme and that scientists should carefully preserve the essential bases of their working condition: free personal exchanges with their colleagues and publication in their scientific journals.

CHRONOLOGICAL TABLE

Sept. 26, 1960 - Identification of 3C 48, the first "quasi-stellar object";

March 1963 - Identification and redshift measurement of 3C 273, the first quasar;

Dec. 12, 1963 - Kardashev submits his original paper;

Mid-Dec., 1963 - Dallas Conference on ultra-high energies;

May 20-23, 1964 - Byurakan Conference on Extraterrestrial Civilizations; Pariiskii identifies CTA 102;

Nov. 27, 1964 - Sandage and Wyndham submit their identification of CTA 102;

Feb. 27, 1965 - Sholomitiskii submits his first results on the variability of CTA 102;

April 8, 1965 - Schmidt submits his measurements of CTA 102 high redshift, making it a quasar;

April 14, 1965 - Pravda announces CTA 102 may be of artificial origin;

May 17, 1965 - Maltby and Moffett submit the first denial of CTA variability;

February 1966 - Sholomitiskii submits his 14-month long variation curve;

1969 - First Soviet scientific monograph on Extraterrestrial Civilizations;

1980 - Milliarcsec resolution radio map of CTA 102.

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